

Schema

version 3 | May 5 2026

Tab delimited Input file: input_data/audits.txt

Emissions Stage encoding - Stage values and integer encoding**

1-7: I=1, II=2, IIIA=3, IIIB=4, IV=5, V=6, {ZE|hybrid|any other} = 7

Emissions rating minimum requirements by policy era and machine group (used in e-bar calculation)

Policy era ending (columns) Machine Groups (rows)	2: 1.9.2020 (+6M) ²				
	1.9.2015 ¹	3: 1.1.2025	4: 1.1.2030	5: 1.1.2040	
Constant speed	IIIA	V	V	V	ZE
CAZ+	IIIB	IV	V	V	ZE
Rest of London	IIIA	IIIB	IV	V	ZE

Non-compliance codes

Non-Compliance Codes come from the multi-values strings used in Initial Machinery Reasons , Final Machinery Reasons , Final Site Reasons , Initial Site Reasons

Code	Description
A	Actively Declined
C	Cannot Evidence Compliance
E	Emissions Standard Not Met
P	Passively Declined
R	Registration Problem
X	[Not Specified]
None	Compliant or reasons not applicable

Rules:

- These are multi-value strings (e.g. `ER`, `AR`, or single `A`), so two boolean flag features are derived: `emissions_compliant` and `admin_compliant`
- If no `E` code present: set `emissions_compliant` = `TRUE`, else `FALSE`
- If any other code present: set `admin_compliant` = `FALSE`, else `TRUE`

Features to retain from `audits.txt`, data type, and first ingestion task

Feature	Type	First ingestion task
Site level data	—	—
<code>Cold-Engaged</code>	chr	<code>Yes/No</code> convert to <code>TRUE/FALSE</code>
<code>Zone</code>		Include where <code>== {"CAZ", "0A", "GL"}</code> ; exclude <code>"P24", "BCP"</code>
<code>Date</code>	Date	yyyy-mm-dd; convert to fractional year float for temporal calculations
<code>Initial Site Compliance</code>	chr	<code>"compliant", "non-compliant", "No NRMM"</code> , check for others
<code>Initial Site Reasons</code>	chr	As List 2, check for other values
<code>Final Site Compliance</code>	chr	As List 2, check for other values
<code>Final Site Reasons</code>	chr	As List 2, check for other values
<code>Audit Num</code>	chr	Visit number to that site
Machine level data	—	—
<code>Initial Machinery Compliance</code>	chr	<code>"emissions compliant", "non-compliant"</code> based on <code>E</code> presence
<code>Initial Machinery Reasons</code>	chr	As List 2, check for other values
<code>Initial Emissions Stage</code>	chr	Encode to int {1-7} per List 1
<code>Initial Retrofit or Exemption</code>	chr	Check for other values
<code>Final Machinery Compliance</code>	chr	As List 2, check for other values
<code>Final Machinery Reasons</code>		As List 2, check for other values
<code>Final Emissions Stage</code>	chr	Encode to int {1-7} per List 1

Feature	Type	First ingestion task
Final Retrofit or Exemption		Check for other values
Machine Type	chr	If No NRMM exclude record
Engine Type	chr	Assign to one of "Constant" , "Variable"
kw Power	chr	If number convert to float; if blank or Unidentified set boolean flag no_power_rating
TAN	chr	Unique identifier for the machine

Equipment groups, in and out of scope

In Scope	Group	Description	Derivation from raw data
Yes	Constant Speed	Generators/constant-speed equipment. Direct IIIA-to-V pathway.	Engine.Type.clean == "Constant" (any Zone)
Yes	CAZ+	Variable-speed machinery in CAZ+. Merges with Rest of London on 1.1.2025.	Engine.Type.clean == "Variable" AND Zone %in% c("CAZ", "0A")
Yes	Rest of London	Variable-speed machinery outside CAZ+.	Engine.Type.clean == "Variable" AND Zone == "GL"
Yes	Variable_Speed	Merged CAZ+ and Rest of London from 1.1.2025. P24 zone only.	Engine.Type.clean == "Variable" AND Zone == "P24"
No	BCP	Beyond Construction Project	Zone == BCP

Rules for compliance outcomes

Row	Site Reg?	NRMM in scope	Mach Reg?	Emissions OK? ³	Enforcement requested	Site mgmt action	Outcome	Emissions reduced by audit?
1	Y	Y	Y	Y	None	None	Self-compliant	No

Row	Site Reg?	NRMM in scope	Mach Reg?	Emissions OK? ³	Enforcement requested	Site mgmt action	Outcome	Emissions reduced by audit?
2	Y/N	Y	N	Y	Register site/machine	Registered	Driven compliant	No
3	Y/N	Y	N	Y	Register site/machine	Not registered	Non-compliant (R)	No
4	Y/N	Y	Y/N	N	Remove or replace	Removed/replaced	Driven compliant	Yes
5	Y/N	Y	Y/N	N	Remove or replace	Not actioned	Non-compliant	No
6	Y/N	N	–	–	None	–	No in-scope plant	No

Model phases and sub-phases

Phase	sub-phase	Start	End	Notes
A				
	A1	1.1.2016	31.12.2018	Pre-Stage V market availability.
	A2	1.1.2019	31.8.2020	Stage V equipment newly available. Machines that received a COVID exemption in the period 1.9.2020–31.3.2021 are included here.
B		1.9.2020	31.12.2024	Tighter LEZ requirements, excluding COVID exempt machines. Phase B subdivided into 3 equal sub-phases of ~527 days to stabilise variance.

Phase	sub-phase	Start	End	Notes
	B1	1.9.2020	10.2.2022	
	B2	11.2.2022	23.7.2023	
	B3	24.7.2023	31.12.2024	
C		1.1.2025	31.12.2030	Forecast period. CAZ+ and Rest of London combined. Initialisation state derived from P24 records (Stage distribution snapshot), not projected from end-2024, using a merged dataset of Constant_Speed and Variable_Speed records.

Modelling approach

The modelling approach is based on Markov modelling,

$$\mathbf{N}^{t+1} = \mathbf{P} \times \mathbf{N}^t$$

where $\mathbf{P}$ is a constant 6x6 transition matrix whose non-zero values are p and $\mathbf{N}^t$ is the vector of stage proportions at time t . This assumes stationary transition probabilities across the entire data period. In each sub-phase, different transition rules apply to give different p , as previous trials found the simple model failed, as the unconstrained 6x6 matrix has 36 free parameter but only 24 observations to constrain it because Stage V accumulates irreversibly.

To address this, separate matrices are estimated for each LEZ phase and sub-group, s:

$$\mathbf{N}^{t+1} = \begin{cases} \mathbf{P}_1 \times \mathbf{N}^t & \text{for } t \in [2016, 2018] (\text{Upgrade to Stage IIIa/IIIb}) \\ \mathbf{P}_2 \times \mathbf{N}^t & \text{for } t \in [2019, 2020] (\text{Upgrade to Stage IIIb/IV/V depending on group}) \\ \mathbf{P}_3 \times \mathbf{N}^t & \text{for } t \in [2021, 2025] (\text{Upgrade to Stage V}) \\ \mathbf{P}_4 \times \mathbf{N}^t & \text{for } t \in [2026, 2030] (\text{Forecast}) \end{cases}$$

Footnotes

1. Policy phase 1 spans 1.9.2015 onwards. 2015 excluded due to lack of audit data. [↩](#)
2. Six-month COVID-19 compliance extension granted. [↩](#)

3. A prior exemption permit satisfies the emissions compliance requirement, but equipment and site must still be registered. [↩](#)